SET I

MORADABAD INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CLASS TEST - 1

Course: B.Tech
Session: 2024-25
Subject: Design and Analysis of Algorithm
Max. Marks: 20

Semester: 5th
Section: A, B, C, D, I, J
Subject Code: BCS-503
Time: 1 Hour

Q. No. :	1	2	3	4	5	6
CO No. :	CO 1	CO 1	CO 1	CO 1	CO 2	CO 1
Bloom's Level	K1, K3	K3	K2	K2, K4	K2, K3	K5

CO1: Design new algorithms, prove them correct, and analyze their asymptotic and absolute runtime and memory demands.
CO2: Find an algorithm to solve the problem (create) and prove that the algorithm solves the problem correctly (validate).

	Section (A) **Attempt all questions.**	**Marks**
1)	Solve the following recurrence using Master method: $T(n)=2T(n/2)+n^3/\log n$	(2)
2)	What do you understand by Asymptotic Notations ?	(3)
3)	Explain Algorithm for Counting Sort.	(3)
	Section (B) **Attempt all questions.**	
4)	Write the Quick Sort Algorithm. Compare the complexity analysis between Quick Sort & Merge Sort.	(4)
5)	Define a Red-Black Tree with its properties. Insert the nodes 10, 18, 7, 15, 16, 30, 25, 40, 60, 2, 1, 70 in empty Red-Black Tree.	(4)
6)	Discuss Max-Heapify and Build-Max-Heap procedures.	(4)

SET I

MORADABAD INSTITUTE OF TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
CLASS TEST - 2

Course: B.Tech
Session: 2024-25
Subject: Design and Analysis of Algorithm
Max. Marks: 20

Semester: 5th
Section: A, B, C, D, I, J
Subject Code: BCS-503
Time: 1 Hour

Q. No. :	1	2	3	4	5	6
CO No. :	5	4	3	3	3	4
Bloom's Level	K2, K3	K2, K4	K2, K5	K2, K5	K2, K5	K2, K4

CO3: Understand the mathematical criterion for deciding whether an algorithm is efficient, and know many practically important problems that do not admit any efficient algorithms.
CO4: Apply classical sorting, searching, optimization and graph algorithms.
CO5: Understand basic techniques for designing algorithms, including the techniques of recursion, divide-and-conquer, and greedy.

		Marks
Section (A) **Attempt all questions.**		
1)	(a) Define any String Matching Algorithm. (b) Explain the concepts of NP Completeness.	(2)
2)	Define N-Queens Problem. Give a Solution for 8-Queens.	(3)
3)	Solve Knapsack Problem using Greedy Approach for the following table. Consider W=5	(3)
Section (B) **Attempt all questions.**		
4)	Construct the minimum spanning tree (MST) for the given graph using Prim's Algorithm-	(4)
5)	Apply Djikstra's Algorithm on given graph for find single source shortest path-	(4)
6)	Apply Floyd Warshall algorithm for constructing all pair shortest path-	(4)

Item	Weight	Value
1	2	3
2	3	4
3	4	5
4	5	6





